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INNOVATION AND TECHNOLOGICAL
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COMMISSION OF EXPERTS
FOR RESEARCH
AND INNOVATION

EFI

REPORT

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EXECUTIVE SUMMARY

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The Commission of Experts wishes to emphasize that the positions expressed in the report do not necessarily represent the opinions of the aforementioned experts.

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Executive Summary

A Current Developments and Challenges

A 0 Where We Stand: Economic Development Weak and R&I System Under Pressure

Germany's economic development is worrying. In terms of growth momentum and research and innovation (R&I), Germany is in most cases lagging far behind the leading major industrialized nations of China, Japan, South Korea and the USA and below the EU average.

The acute economic weakness is accompanied by a profound structural weakness. Among other things, the effects induced by digitalization and decarbonization are insufficiently matched by innovations and new business models that could generate economic momentum and new jobs. In terms of economic policy, therefore, traditional economic stimulus programmes are not indicated. Rather, the main aim must be to utilize the revitalizing power of structural change.

With a well-positioned, internationally leading R&I system, Germany could ignite a fresh drive for growth. The resulting economic strength would also significantly improve the chances of mastering the upcoming major transformations in digitalization and decarbonization and securing the technological and economic sovereignty of Germany and Europe. For this reason, the future Federal Government should place greater focus on the entire R&I system and its performance in its R&I policy strategy, in addition to various priority areas.

A 1 20th Legislative Period: R&I Policy Nowhere Near the Proclaimed 'German Pace'

In view of the grand societal challenges, the Federal Government of the 20th legislative period has committed itself to the goal of driving forward economic and social transformation through digitalization and decarbonization. In doing so, it left no doubt that research and innovation (R&I) and, specifically, a reorientation of German R&I policy are of vital importance for achieving this goal.

However, the implementation of such a reorganized R&I policy was hampered by a fundamental deficit: slowness. This slowness results equally from prioritization and coordination deficits as well as implementation deficits. The Federal Government's R&I policy is therefore nowhere near the 'German pace' it has proclaimed. Many of the R&I policy projects announced in the coalition agreement were not implemented. In addition, the Federal Government did not succeed in significantly advancing the expansion of transformation infrastructures and reducing regulatory barriers to innovation to a sufficient extent.

The Commission of Experts comes to the sobering conclusion that, despite its high R&I policy ambitions, the Federal Government has not succeeded in sparking the innovation drive necessary for the transformations.

A 2 21st Legislative Period: Making R&I Policy More Impactful

An efficient R&I system is a key factor for the development of internationally competitive companies, for the transformation towards an environmentally friendly economy, for sovereignty in key enabling and future technologies and for the digital transformation. The German R&I system currently does not adequately fulfil these demands. R&I policy in the coming legislative period is therefore required to take measures to strengthen this system.

Although R&I policy has already taken on this task in recent years, it lacks impact in many areas. How can this change in the coming legislative period? The Commission of Experts considers it necessary to further develop the policy approach of the New Mission Orientation, to establish adequate governance structures for R&I policy, to take a closer look at the effectiveness and efficiency of policy measures and to create general conditions that are more conducive to innovation.

A 3 Industrial Policy

Industrial policy has been experiencing a renaissance in recent years. Industrial policy measures are primarily taken to promote sustainability, boost the competitiveness of German and European companies and ensure technological and economic sovereignty. As their use is associated with risks, the following aspects should be taken into account:

Industrial policy measures should not be taken to compensate for shortcomings in other policy areas. They cannot replace innovation-friendly regulatory and institutional framework conditions but merely supplement these.

If vertical industrial policy measures are used in addition to horizontal measures, they should be primarily focused on potentially high-growth and research-intensive sectors to ensure long-term success and generate spillover effects.

Vertical industrial policy measures should only fulfil a catalytic function and not provide companies with long-term support. As with the New Mission Orientation, the aim is to create a stimulus and then to discontinue support.

Good industrial policy is characterized by the fact that it promotes entrepreneurial activity. It should primarily facilitate the creation and growth of new companies and largely hold back on supporting established companies.

B Core Topics 2025

B 1 Transformative Structural Change Through Digitalization and Decarbonization

Economic growth and structural change are closely linked in transformation processes. Two facets of structural change are currently the focus of attention in Germany: digitalization is boosting the competitiveness of companies; the shift to a carbon-neutral economy is key in the transition towards an environmentally sustainable economy. The analyses in this chapter indicate that Germany is lagging far behind in developing and implementing new digital products, processes and business ideas. This can be remedied by expanding digital infrastructures and improving the legal framework for data access and exchange. To make better use of the remaining potential for decarbonization, climate protection measures must be designed for efficiency and stronger incentives for innovation must be created.

Instead of preserving existing structures, policymakers should actively support the transformative structural change associated with digitalization and decarbonization. Given the high percentage of employees who work in occupations with high digitalization potential or in environmentally harmful occupations and face significant pressure to adapt, particular emphasis should be placed on measures promoting labour market mobility and employee qualification.

The Commission of Experts recommends the following measures, among others:

- Research and innovation (R&I), technology transfer between science and industry as well as start-ups should be widely promoted in order to generate potential for creating new jobs in good time.
- The fibre optic network is essential for the competitiveness and innovative capacity of companies and should be expanded rapidly, particularly in rural regions. In addition, the technical and legal conditions for optimal data access and data linking should be created in order to open up opportunities for digitalization.

- Climate policy measures should be designed efficiently to incentivize companies to develop environmentally friendly technologies, in particular by means of a standardized carbon price across all sectors. Carbon contracts for difference can support innovative production processes if sufficient competition for funding helps to avoid excessive government cost burdens.

- Regional policy measures should be geared towards supporting transformative structural change, for example by coordinating measures across government departments and promoting innovation ecosystems through collaborative projects.

- Employee mobility and continuing education should be promoted. In this regard, the Federal Employment Agency's support instruments should in future prepare employees in occupations particularly affected by automation or the discontinuation of environmentally harmful activities more intensively for transfers to other sectors.

B 2 Quantum Technologies

New quantum technologies harbour great potential for innovation and are regarded as key enabling technologies of the future. Quantum computing promises a significant increase in computing power and could significantly accelerate the solution of highly complex optimization problems or even make them possible for the first time. At the same time, this makes protection against cyberattacks increasingly urgent, which requires the development of secure encryption technologies in the long term – an area in which quantum communication offers promising approaches. Quantum sensor technology significantly improves the sensitivity and precision of measurement and imaging technology.

Many of these new quantum technologies are still in the early stages of development. Germany is in a strong starting position, which must be safeguarded in co-operation with its EU partners. This will ensure that German companies remain competitive in global competition with the USA and China and that Germany is technologically sovereign and well prepared for security threats in the long term.

The Commission of Experts therefore recommends the following measures, among others:

- With a national quantum strategy, the Federal Government should create a coherent framework for the further development of quantum technologies. To justify long-term investments in ongoing and pioneering projects, it should promptly develop an ambitious follow-up concept for the Quantum Technologies Conceptual Framework Programme, which expires in 2026, and swiftly implement it.

- The establishment of an R&I quantum ecosystem should be consistently conceptualized on a European level. Among other things, national competences and efforts in quantum computing should be pooled across Europe to facilitate quantum computing 'Made in Europe' and avoid dependencies on large non-European companies in the medium and long term.
- Research and development activities in quantum technologies require well-connected actors. At national level, the existing regional R&I clusters should therefore be expanded and integrated into a national strategy. The networking between these clusters should be actively supported, synergy potential between civil and military R&D should be utilized and a broad range of actors across regions should be given low-threshold access to the research and computing infrastructure.
- The transfer of findings from basic and application-oriented quantum research into innovative, marketable products should be actively promoted through government anchor customer contracts, the creation of a supportive environment for quantum start-ups and measures to strengthen the venture capital market.

B 3 Innovation in the Water Industry

Water is the source of life for humans, animals and plants. Although Germany is a water-rich country, climate change will lead to more frequent regional and seasonal water shortages in the future, resulting in conflicts of use among and between private, commercial and agricultural consumers. Moreover, water infrastructures must be adapted to more frequent extreme weather events such as droughts and heavy rainfall. Finally, water quality is affected by the infiltration of fertilisers and pesticides from agriculture into the groundwater and by pharmaceutical residues and microplastics, which are still not sufficiently removed from wastewater.

Water management innovations can help to meet these challenges. Compared internationally, Germany is in a strong position here. However, the adoption of these innovations faces numerous obstacles in the domestic water industry, which arise, among other things, from its high level of fragmentation, considerable risk aversion and a strong focus on compliance with prescribed technical standards.

The Commission of Experts therefore recommends the following measures, among others:

— The Commission of Experts supports the testing of innovative concepts in regulatory sandboxes as envisaged in the National Water Strategy. In addition to technical solutions, institutional innovations such as the adaptation of water withdrawal rights and water trading should be trialled there.

— Sustainable water management requires data on allocated water rights and actual water withdrawals. The recording in a comprehensive and transparent water register, as announced in the National Water Strategy, must be implemented swiftly and in digital form.

— Water withdrawal charges must be adjusted in such a way that they reflect the scarcity of water, particularly in times of prolonged drought, and thus the external costs of water withdrawal. The extended producer responsibility of industry should be quickly translated into national law and implemented.

— The measures planned by the Federal Government to promote alternative forms of financing should be implemented swiftly to meet the special needs of social enterprises.

— The use of pesticides and fertilisers should be subject to a levy, following the example of Denmark, so that polluters reduce water pollution.

— To create stronger incentives for the adoption and development of innovations, concepts and measures should be drafted that make it more attractive for smaller supply units to merge into larger supply units.

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